DESCENT OF HOM GROUPS OUT OF A PSEUDO-COHERENT OBJECT INTO A RELATIVELY FINITE TOR DIMENSION OBJECT ALONG A FILTERED LIMIT OF SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Mayer-Vietoris for Hom in the derived category: Hom out of a compact-free source can be computed across unions of opens, so a local isomorphism statement propagates from U_0, V_0 and U_0 cap V_0 to U_0 cup V_0. invoked at proof lines 9082-9085. Provides the gluing step of the local condition P.
- The induction principle for quasi-compact opens, applied first to opens of S_0 and then to quasi-compact quasi-separated opens of X_0. invoked at proof lines 9090-9098. Reduces the assertion to the affine-affine case.
- In the affine case the derived-category Hom statement translates into module Hom computations, with K_0 pseudo-coherent and L_0 of finite tor dimension over the ring, by the pseudo-coherence, affine-compare, pseudo-coherent-affine, and tor-dimension lemmas. invoked at proof lines 9101-9112. Sets up the hypotheses of the More on Algebra colimit lemma for relatively perfect systems.

Source reference. Original remark 0DI6: sources/stacks-chow/tags.txt, source lines 16931–16931. Cited at proof line 9058 as the standing situation of the lemma; the gluing and reduction arguments are inside the counted span.

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 10581–10581. Cited at proof lines 9084-9085 to glue the condition P over unions; the gluing argument is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Cited at proof lines 9091-9092 and used again at proof lines 9096-9098 to reduce to the affine-affine case; both reductions are inside the counted span.

Source reference. Original lemma 0DHX: sources/stacks-chow/tags.txt, source lines 16922–16922. Cited at proof lines 9102 and 9114 to conclude in the affine case; the translation into module terms is inside the counted span.

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Source reference. Original lemma 08E5: sources/stacks-perfect/pinned-original-source.tex, source lines 2422–2422. Cited at proof line 9103 to translate the statement; the module-level translation is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 9103-9104 to translate Homs into module Homs; the translation is inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 9107-9109 for pseudo-coherence of the module complex of K_0; the affine case assembly is inside the counted span.

Source reference. Original lemma 0DHY: sources/stacks-perfect/pinned-original-source.tex, source lines 2552–2552. Cited at proof lines 9110-9112 for the finite tor dimension of the module complex of L_0; the affine case assembly is inside the counted span.

Lemma 0.2. *In Situation 0DI6. Let K_0 and L_0 be objects of $D(\mathcal{O}_{X_0})$. Set $K_i = Lf_{i0}^*K_0$ and $L_i = Lf_{i0}^*L_0$ for $i \geq 0$ and set $K = Lf_0^*K_0$ and $L = Lf_0^*L_0$. Then the map*

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{X_i})}(K_i, L_i) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_X)}(K, L)$$

is an isomorphism if K_0 is pseudo-coherent and $L_0 \in D_{QCoh}(\mathcal{O}_{X_0})$ has (locally) finite tor dimension as an object of $D((X_0 \to S_0)^{-1}\mathcal{O}_{S_0})$

Proof. For every quasi-compact open $U_0 \subset X_0$ consider the condition P that

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{U_i})}(K_i|_{U_i}, L_i|_{U_i}) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_U)}(K|_U, L|_U)$$

is an isomorphism where $U = f_0^{-1}(U_0)$ and $U_i = f_{i0}^{-1}(U_0)$. If P holds for U_0, V_0 and $U_0 \cap V_0$, then it holds for $U_0 \cup V_0$ by Mayer-Vietoris for hom in the derived category, see Cohomology, Lemma 08BW.

Denote $\pi_0 : X_0 \to S_0$ the given morphism. Then we can first consider $U_0 = \pi_0^{-1}(W_0)$ with $W_0 \subset S_0$ quasi-compact open. By the induction principle of Cohomology of Schemes, Lemma 08DR applied to quasi-compact opens of S_0 and the remark above, we find that it is enough to prove P for $U_0 = \pi_0^{-1}(W_0)$ with W_0 affine. In other words, we have reduced to the case where S_0 is affine. Next, we apply the induction principle again, this time to all quasi-compact and quasi-separated opens of X_0, to reduce to the case where X_0 is affine as well.

If X_0 and S_0 are affine, the result follows from More on Algebra, Lemma 0DHX. Namely, by Lemmas 08E5 and 06Z0 the statement is translated into computations of homs in the derived categories of modules. Then Lemma 08E7 shows that the complex of modules corresponding to K_0 is pseudo-coherent. And Lemma 0DHY shows that the complex of modules corresponding to L_0 has finite tor dimension over $\mathcal{O}_{S_0}(S_0)$. Thus the assumptions of More on Algebra, Lemma 0DHX are satisfied and we win. □